



SECR1033 COMPUTER ORGANIZATION & ARCHITECTURE (2023/2024-2)

GROUP PROJECT

CPU & Hardware Performance Benchmarks

(PART 1)

CPU Benchmarking and Performance Analysis Using Free Tools

Introduction : In computing, a benchmark is the act of running a computer program, a set of programs, or other operations, in order to assess the relative performance of an object, normally by running a number of standard tests and trials against it.

A CPU benchmark (CPU benchmarking) is a series of tests designed to measure the performance of a computer or device CPU. A set of standards, or baseline measurements are used to compare the performance of different systems, using the same methods and circumstances.

There are available CPU benchmark software available and free for download. We may explore, try and experience it on our own computer.

-
- Instructions** :
1. Go through each video in additional document provided;
Attachment doc
 2. Select 3 benchmarking tools to be used in your project;
 3. Select 3 different computers / notebook in your team to be benchmarked (Windows OS);
 4. Perform the benchmarking on those computers;
 5. Compare and analyze the performance;
 6. Write a comprehensive report for each group (**70 Marks**);
 7. Provide a 10 – 15 minutes video presenting your group project (each member **MUST** presents with camera on) (**30 Marks**)

PART 1: CPU Benchmarking and Performance Analysis Using Free Tools

Motivation : Exploring CPU benchmarking can be an exciting project for students, offering insights into computer performance, hardware differences, and the intricacies of software optimization.

Objectives :

1. Understand the principles of CPU benchmarking.
2. Compare the performance of different CPUs.
3. Analyze the impact of different workloads on CPU performance.
4. Learn to use free benchmarking tools and interpret their results.

Tools : Use the following freeware benchmarking software to conduct a performance benchmarking to one (1) computer/laptop.

- **CPU-Z**: For detailed CPU information and simple benchmarking. Passive attack to get basic System Info.
- **Geekbench**: For cross-platform CPU benchmarking. CPU stress test.
- **Windows Task Manager**: press `Ctrl + Alt + Delete` and select "Task Manager" from the options screen. To explore load of running apps and computer resources.
- **MiniTool Partition Wizard Free 12.8** - Disk Benchmark

Steps :**1. Introduction to CPU Benchmarking:**

- Research and present the basics of CPU benchmarking.
- Explain different types of benchmarks (synthetic, real-world, and application benchmarks).

2. Setup and Installation:

- Provide instructions on how to download and install each benchmarking tool.
- Ensure each tool is properly configured for accurate results.

3. Benchmarking Process:

- Use CPU-Z for collecting spec of your computer (Laptop)
- Use Geekbench for running CPU stress test (CPU test)
- Use Task Manager (Windows OS Tools) to simulate Impact of Different Workloads (Task Manager – Windows OS Tool)
 - High CPU load (e.g., rendering a video).
 - Mixed load (e.g., gaming while running background processes).
 - Low CPU load (e.g., basic office tasks).
- Analyze how different workloads will affect CPU resources and performance
- Use MiniTool Partition Wizard to perform Disk Benchmarking
 - Do Read and Write stress test to External Disk (HDD or SDD)

4. Data Collection and Analysis:

- Collect benchmarking process at least two (2) different computer (or laptop)
- Compile the benchmarking results in a structured format (*e.g.* in a comparison table).
- Analyze the data to compare CPU performance across different benchmarks.
- Create graphs and charts to visually represent the data.

5. Report Writing:

- Summarize the findings in a detailed report.
- Discuss the strengths and weaknesses of each CPU based on the benchmarking results.
- Include visual aids (graphs, charts) to support the analysis.

6. Presentation:

- Prepare a presentation to share the findings with peers or instructors.
- Highlight key insights and takeaways from the benchmarking study.

- Expected Outcomes :**
- A comprehensive understanding of CPU performance metrics.
 - Hands-on experience with benchmarking tools.
 - Ability to analyze and interpret benchmarking data.
 - Enhanced research and presentation skills.

- Resources :**
- **CPU-Z** (<https://www.cpubid.com/software/cpu-z.html>)
 - **Geekbench** (<https://www.geekbench.com/>)
 - Windows Task Manager (internal tools provide by Windows OS)
 - MiniTool Partition Wizard Free 12.8
<https://www.partitionwizard.com/free-partition-manager.html>
-

Activities and Case study to be considered:

Benchmark a Computer/Laptop

1. Use CPU-Z for collecting spec of your computer (Laptop)
 - CPU processor (clock frequency/speed)
 - Cache
 - Memory
 - Mother board
2. Use Geekbench
 - Add system info as complement /conformation to CPU-Z
 - CPU test with variation of technique or method used, either singlecore or multi-core performance.
3. Monitor Task Manager (Windows OS tools)
 - Invoke different process and application as categorised as low, middle, high load.
Monitor the following
 - Monitor process
 - Monitor performance
 - Low Load
 - Open Word Editing (Microsoft Word processing) as memory consumption is around 30% - 50% (example)
 - Middle Load

- Open few application such as memory consumption is around 50% to 60% (example)
 - High Load
 - Open more application such as memory consumption above 70% (example)
 - e.g. Microsoft Visual Studio, Mathlab, Alteryx
 - Games
 - Steaming
 - If the load is almost full i.e. memory consumption is above 90%, computer performance is reduced. By killing the highest and unwanted process, it will release more free memory and as a result memory consumption will be released.
 - Do some activities to show memory consumption can be released and looking at the Monitor performance.
4. Use Use MiniTool Partition Wizard (Windows) to perform Disk Benchmarking process
- Do Stress test for Read and Write to your External Disk (HDD or SDD)
 - Use different mode test: Sequential, and Random
5. Each student is needed to do benchmark of his/her computer/laptop by executing steps 1 to 4 above. Record your result in a table and arrange the result in a proper and suitable manner. Example as below.

Result Example: Benchmark for computer/laptop

Computer	CPUZ (System Info)	Geekbench (CPU stress test)	MiniTools (Disk benchmark)	Task Manager (Active Program and Resource Adjustment)
A				

6. Compare benchmark for different computers using activities 1 – 4. Outline the result in a table and discuss the result. Example as below.

Result Example: Benchmark Comparison

Computer	CPUZ (System Info)	Geekbench (CPU stress test)	MiniTools (Disk benchmark)	Task Manager (Active Program and Resource Adjustment)
A				
B				
C				
D				



SECR1033 COMPUTER ORGANIZATION & ARCHITECTURE (2023/2024-2)

GROUP PROJECT

CPU & Hardware Performance Benchmarks

(PART 2)

Writing ASM code program for Benchmarking software

Introduction : Write an assembly language program to calculate and measure the time taken to execute different loops for the polynomial equation:

$$y = (a * x^3) + (b * x^2) + c * x + d$$

where a , b , c , d are coefficients delay and x is an integer input values. Implement a loop that iterates different loop times, each time calculating the value of y for a different x value.

Different loops will represent different load that will impose to the polynomial equation as benchmarking method. Measure the time taken to complete the different loop and output the result in **Table 1** for comparison purpose. Assume that the input x values are consecutive integers starting from 1 to value of max defined looping N . Use appropriate timing instructions or techniques to accurately measure the execution time.

Instructions

1. Use polynomial equation as benchmarking algorithm:

$$y = (a * x^3) + (b * x^2) + c * x + d$$

where a , b , c , d are coefficients delay value and x is an input integer value

$$1 < x < \text{max_loop}(N)$$

2. Setting delay coefficient a , b , c , and d :

- If you have 4 students in a group, select your date (Day) in your birthday to set value a , b , c , and d .

- *Example:*
 - Student 1, 2, 3, and 4 birthday are **05-03-2000**, **08-03-2000**, **15-05-2000**, **10-07-2000**
 - Then coefficient $a = 05$, $b = 08$, $c = 15$, $d = 10$
 - If you have only 3 students, value d you choose any random value.

3. Use the defined coefficient and execute math calculation for benchmarking, use the following pseudo code:

```

max = max_loop (N); e.g. N=100
sum = 0;
coef1 = 5;
coef2 = 8;
coef3 = 15;
coef4 = 10;

for (int i = 1; i <= max; i++)
{
    y = (coef1 * x^3) + (coef2 * x^2) + coef3*x + coef4;
    sum = sum + y
}

```

Note: coef1,coef2,coef3 and coef4 are delay factor process in msec

4. Capture CPU elapse time, use *call GetMseconds*

- *Before start math loop calculation*
- *After complete math loop calculation*
- *CPU elapse time = different time before and after run math equation in loop*

Use irvine library (call PROC): refer  [Irvine Library](#)

GetMseconds PROC

Returns the number of milliseconds that have elapsed since midnight.

Call args: None

Return arg: EAX = ((hours*3600)+(minutes*60)+ (seconds))*1000 + milliseconds

Example `call` GetMseconds:

```
.data

capture_msec DWORD ?

.code

call GetMseconds
mov capture_msec, eax ;save the capture time in msec

; example of coding c*x in y=(a*x^3)+(b*x^2)+c*x+d

mov eax, coef3 ; delay factor coef3
call Delay
mul bx
add sum, ax ; partial sum for coef3
```

5. Display the output:

- Algorithm used, equation y and loop
- CPU elapse time

$$\text{CPU elapse time} = \text{capture_msec (after)} - \text{capture_msec (before) loop}$$

6. Run different `max_loop` (N) value to compare performance difference (e.g. 50, 100, 200, 500, 700, 1000)

Reference:

<https://csc.csudh.edu/mmccullough/asm/help/>

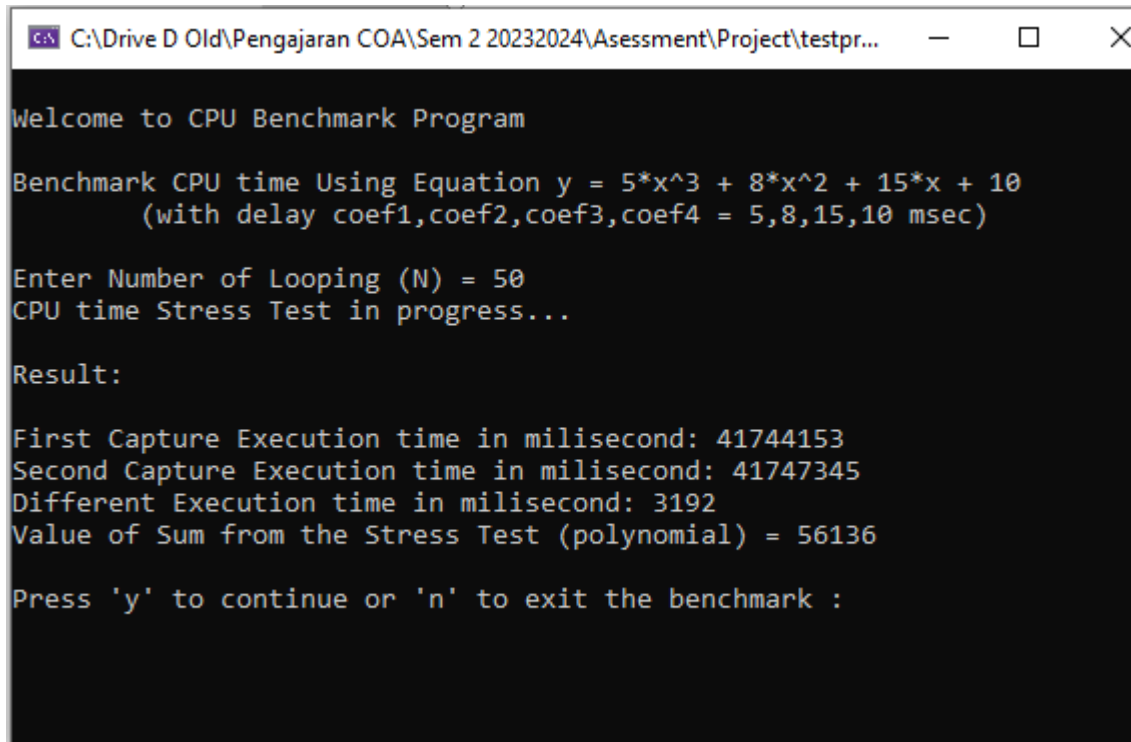


[Irvine Library](#)

Explore under Irvine Library list – there are various available proc asm that can be used by user by using method `call` in .main asm code.

Some invine library (proc) might useful for your asm coding:

- `call` Clrscr, `call` Crlf, `call` WriteString, `call` ReadDec, `call` GetMseconds, `call` WriteDec, `call` Delay, `call` ReadChar, `call` WriteChar

Example of Output:

```
C:\Drive D Old\Pengajaran COA\Sem 2 20232024\Assessment\Project\testpr...

Welcome to CPU Benchmark Program

Benchmark CPU time Using Equation  $y = 5x^3 + 8x^2 + 15x + 10$ 
(with delay coef1,coef2,coef3,coef4 = 5,8,15,10 msec)

Enter Number of Looping (N) = 50
CPU time Stress Test in progress...

Result:

First Capture Execution time in milisecond: 41744153
Second Capture Execution time in milisecond: 41747345
Different Execution time in milisecond: 3192
Value of Sum from the Stress Test (polynomial) = 56136

Press 'y' to continue or 'n' to exit the benchmark :
```

Table 1 : Comparison Table

No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	41744153	41747345	3192
100			
200			
500			
700			
1000			

Report format :

Section	Topic	Marks
	Cover	2
	TOC (Table of Content)	
1.0	Introduction - Problem Background, aim, objective, scopes (computer specification & benchmarking tools features)	10
2.0	The flowchart of the project (benchmarking methodology / procedure)	10
3.0	Benchmarking results	15
4.0	Analysis, comparison, discussion	15
5.0	Conclusion and reflection	10
	References	4
	Appendices (Tasks distribution, group pictures)	4
TOTAL MARKS :		70

Video format :

Contents	Marks
Introduction of group members	2
Brief Introduction - Problem Background, aim, objective, scopes (computer specification & benchmarking tools features)	5
Benchmarking execution and results	7
Comparison and discussion	7
Conclusion	6
Reflection (each member)	3
TOTAL MARKS :	30

Submission :

Deadline	Open submission From (revision week until Final Examination Date (15 July 2024))												
Documents/materials (submitted to e-learning)	1. Hardcopy of the report 2. Link of the video (on the front page of report) • Demo on how benchmarking of CPU by using Freeware tools (short video). Example to one (1) computer only. • Demo on running CPU benchmarking using your own ASM code (short video) • Presentation of final project group (slide presentation) (short video) 3. Project folder (zip folder) <table><tr><th>Name ^</th><th>Date modified</th><th>Type</th></tr><tr><td> Debug</td><td>10/6/2024 11:30 AM</td><td>File folder</td></tr><tr><td> testproject</td><td>10/6/2024 12:43 PM</td><td>File folder</td></tr><tr><td> testproject.sln</td><td>24/6/2023 9:49 PM</td><td>Visual Studio Sol...</td></tr></table>	Name ^	Date modified	Type	 Debug	10/6/2024 11:30 AM	File folder	 testproject	10/6/2024 12:43 PM	File folder	 testproject.sln	24/6/2023 9:49 PM	Visual Studio Sol...
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Prepared by:

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APPENDIX A

MUST SEE VIDEOS

Several interesting videos can be found as follow. You may go through all the videos one by one for better understanding about the benchmarking a computer performance. Hope these videos help you to learn, understand, and gain experience about benchmarking.

1	Title : Benchmarks as Fast As Possible.
	Link : https://youtu.be/cHtNJngtS1I
	Remark : A good introduction for benchmarks.

2	Title : Best CPU Benchmark Software For Windows.
	Link : https://youtu.be/fgWGtWsDXmw
	Remark : You may download the CPU-Z for a start, look through at your CPU and hardware specification.

3	Title : First 5 Things I Do When Benchmarking.
	Link : https://youtu.be/19eUZ0GwTxI
	Remark : Gaming! Easy to relate right. However, benchmark is not limited to GPU, in our case we focus on CPU and hardware performance benchmarking.

4	Title : What's the Difference Between a CPU and a GPU?
	Link : https://blogs.nvidia.com/blog/2009/12/16/whats-the-difference-between-a-cpu-and-a-gpu/
	Remark : This is extra information.

Credited to: Mrs Rashidah Binti Kadir

APPENDIX B

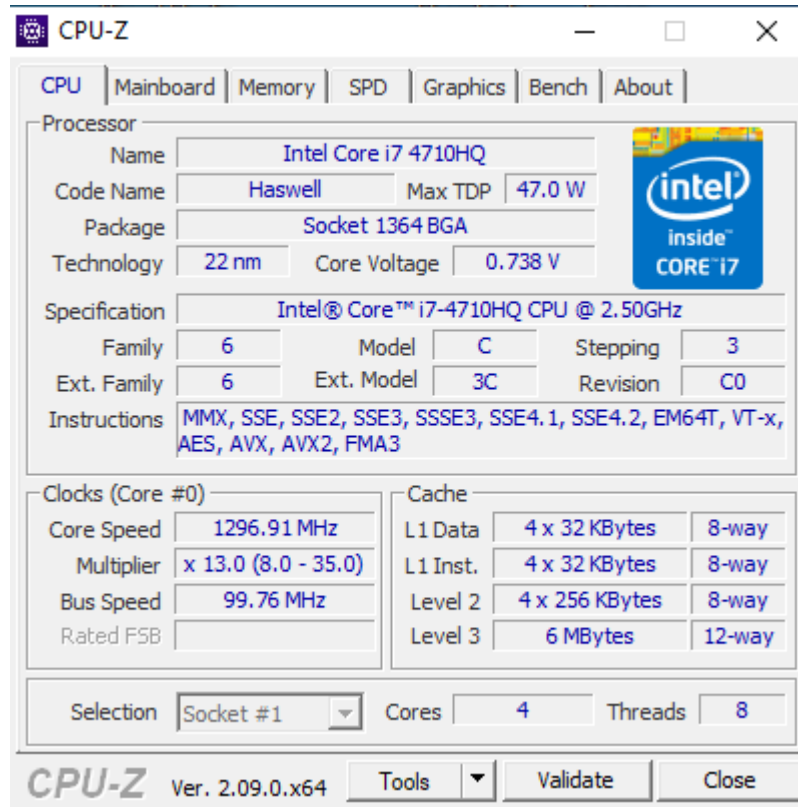


Figure A1: Example of CPU-Z output

The screenshot shows the Geekbench 6 application window. The 'System Information' tab is selected. The operating system is Microsoft Windows 10 Home Single Language (64-bit). The model is ASUSTeK COMPUTER INC. G551JM. The motherboard is ASUSTeK COMPUTER INC. G551JM. The memory is 8.00 GB DDR3 SDRAM. The CPU information shows: Name: Intel Core i7-4710HQ, 1 Processor, 4 Cores, 8 Threads; Codename: Haswell; Package: Socket 1364 BGA; Base Frequency: 2.49 GHz; Maximum Frequency: 3491 MHz. A 'Running CPU Benchmark' dialog box is open, showing a progress bar and a 'Cancel' button. The CPU benchmark is running, with 00:01:01 elapsed. The GPU benchmark is also visible, showing GPU API: OpenCL and GPU: NVIDIA GeForce GTX 860M.

CPU Information	
Cluster 1	4 Cores
Maximum Frequency	3491 MHz
Package	Socket 1364 BGA
Codename	Haswell
L1 Instruction Cache	32.0 KB x 4
L1 Data Cache	32.0 KB x 4
L2 Cache	256 KB x 4
L3 Cache	6.00 MB x 1
Memory Information	
Size	8.00 GB
Transfer Rate	1596 MT/s
Type	DDR3 SDRAM
Channels	2

Figure A2: Example of Geekbench output

Single-Core Performance		Multi-Core Performance	
Single-Core Score	954	Multi-Core Score	2813
File Compression	932 133.8 MB/sec	File Compression	1785 256.3 MB/sec
Navigation	1105 6.66 routes/sec	Navigation	4378 26.4 routes/sec
HTML5 Browser	1043 21.4 pages/sec	HTML5 Browser	3339 68.3 pages/sec
PDF Renderer	1039 24.0 Mpixels/sec	PDF Renderer	4120 95.0 Mpixels/sec
Photo Library	805 10.9 images/sec	Photo Library	2911 39.5 images/sec
Clang	1087 5.35 Klines/sec	Clang	3833 18.9 Klines/sec
Text Processing	859 68.8 pages/sec	Text Processing	1015 81.3 pages/sec
Asset Compression	1004 31.1 MB/sec	Asset Compression	3267 101.2 MB/sec
Object Detection	398 11.9 images/sec	Object Detection	1126 33.7 images/sec
Background Blur	1240 5.13 images/sec	Background Blur	3045 12.6 images/sec
Horizon Detection	1464 45.6 Mpixels/sec	Horizon Detection	4728 147.1 Mpixels/sec
Object Remover	763 58.7 Mpixels/sec	Object Remover	2612 200.8 Mpixels/sec
HDR	1030 30.2 Mpixels/sec	HDR	3057 89.7 Mpixels/sec
Photo Filter	1176 11.7 images/sec	Photo Filter	3000 29.8 images/sec

Figure A3: CPU Load Stress Test (CPU Test)

APPENDIX C

Task Manager:

Windows Task Manager is a system monitoring utility included with the Microsoft Windows operating systems. It provides detailed information about the computer's performance and running applications, processes, and services. It also allows users to manage these processes and troubleshoot issues. Here are some key features of the Windows Task Manager:

1. **Processes Tab:** Lists all the running processes and their resource usage, including CPU, memory, disk, network, and GPU usage. Users can end tasks, set priorities, and see detailed information about each process.
2. **Performance Tab:** Displays real-time graphs and statistics on the system's performance, including CPU, memory, disk, Ethernet, and GPU usage. It provides insights into how system resources are being utilized.
3. **App History Tab:** Shows resource usage information for applications over time, which can be useful for identifying which apps consume the most resources.
4. **Startup Tab:** Lists all the programs that start automatically with Windows. Users can enable or disable these startup programs to improve boot times and overall system performance.
5. **Users Tab:** Displays information about users currently logged into the computer, along with their resource usage. This is helpful in a multi-user environment for identifying who is consuming resources.
6. **Details Tab:** Provides a more detailed view of the processes running on the system, similar to the Processes tab but with additional information and options like setting CPU affinity and analyzing wait chains.
7. **Services Tab:** Lists all the Windows services, along with their status (running or stopped). Users can start, stop, and manage services from this tab.

An output sample of moderate to high load applications, which are running is illustrated in Figure A4.

The screenshot shows the Windows Task Manager Performance tab. The 'Performance' tab is selected, displaying a table of running applications and their resource usage. The table has columns for Name, Status, PID, Process name, CPU, Memory, Disk, Network, GPU, and GPU engine. The 'Apps (8)' section lists Geekbench 6, Google Chrome (16), Microsoft Edge (17), Microsoft Word, Snipping Tool, Task Manager, TradingView (18), and Windows Explorer. The 'Background processes (93)' section lists various Adobe services and background processes. The CPU usage is 11%, Memory is 77%, Disk is 27%, Network is 0%, and GPU is 2%.

Name	Status	PID	Process name	11% CPU	77% Memory	27% Disk	0% Network	2% GPU	GPU engine
Apps (8)									
Geekbench 6		20780	Geekbench 6.exe	0%	39.3 MB	0 MB/s	0 Mbps	0%	
Google Chrome (16)				0.3%	632.3 MB	0.1 MB/s	0 Mbps	0%	GPU 0 - 3D
Microsoft Edge (17)				0.1%	484.5 MB	0 MB/s	0 Mbps	0%	GPU 0 - 3D
Microsoft Word		8800	WINWORD.EXE	0.2%	116.3 MB	0 MB/s	0 Mbps	0%	
Snipping Tool		6860	SnippingTool.exe	0.6%	3.1 MB	0 MB/s	0 Mbps	0%	
Task Manager		20248	Taskmgr.exe	0.4%	26.0 MB	0 MB/s	0 Mbps	0%	
TradingView (18)				1.0%	904.0 MB	0.1 MB/s	0.1 Mbps	0%	GPU 0 - 3D
Windows Explorer		2908	explorer.exe	0.4%	46.0 MB	0 MB/s	0 Mbps	0%	
Background processes (93)									
Acrobat Collaboration Synchro...		14488	AdobeCollabSync.exe	0.1%	2.9 MB	0 MB/s	0 Mbps	0%	
Acrobat Collaboration Synchro...		7392	AdobeCollabSync.exe	0%	1.3 MB	0 MB/s	0 Mbps	0%	
Acrobat Licensing Service (32 bit)		15876	acrotray.exe	0%	2.9 MB	0 MB/s	0 Mbps	0%	
Acrobat Update Service (32 bit)		4244	armsvc.exe	0%	0.1 MB	0 MB/s	0 Mbps	0%	
Adobe Content Synchronizer (3...		14384	CoreSync.exe	0%	13.1 MB	0 MB/s	0 Mbps	0%	
Adobe Crash Processor (32 bit)		12816	Adobe Crash Processor.exe	0%	1.2 MB	0 MB/s	0 Mbps	0%	
Adobe Genuine Software Monit...		4768	AGMSvc.exe	0%	1.5 MB	0 MB/s	0 Mbps	0%	
Agent for EasyConnect (32 bit)		19260	ECAgent.exe	0.1%	2.2 MB	0 MB/s	0 Mbps	0%	

Figure A4: Process versus resources usage (CPU, Memory, Disk, Network, GPU)

APPENDIX D

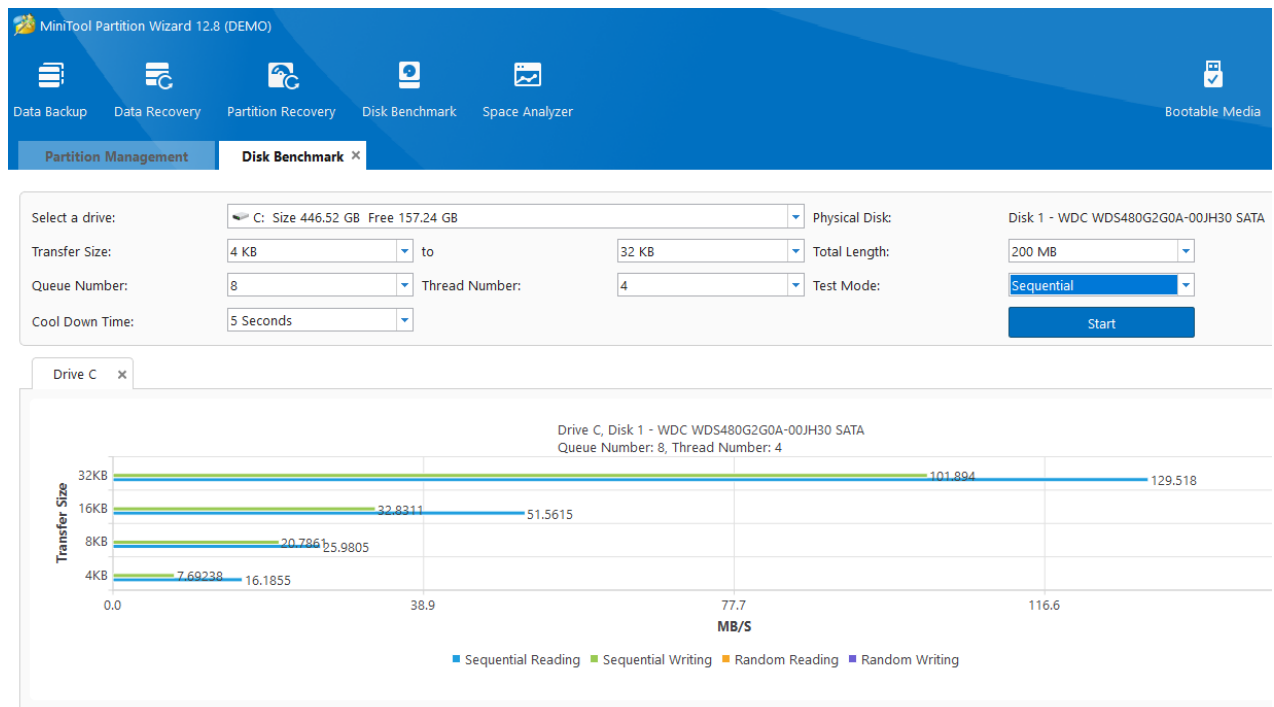


Figure A5: Example of Disk Benchmark (Test Mode: Sequential)

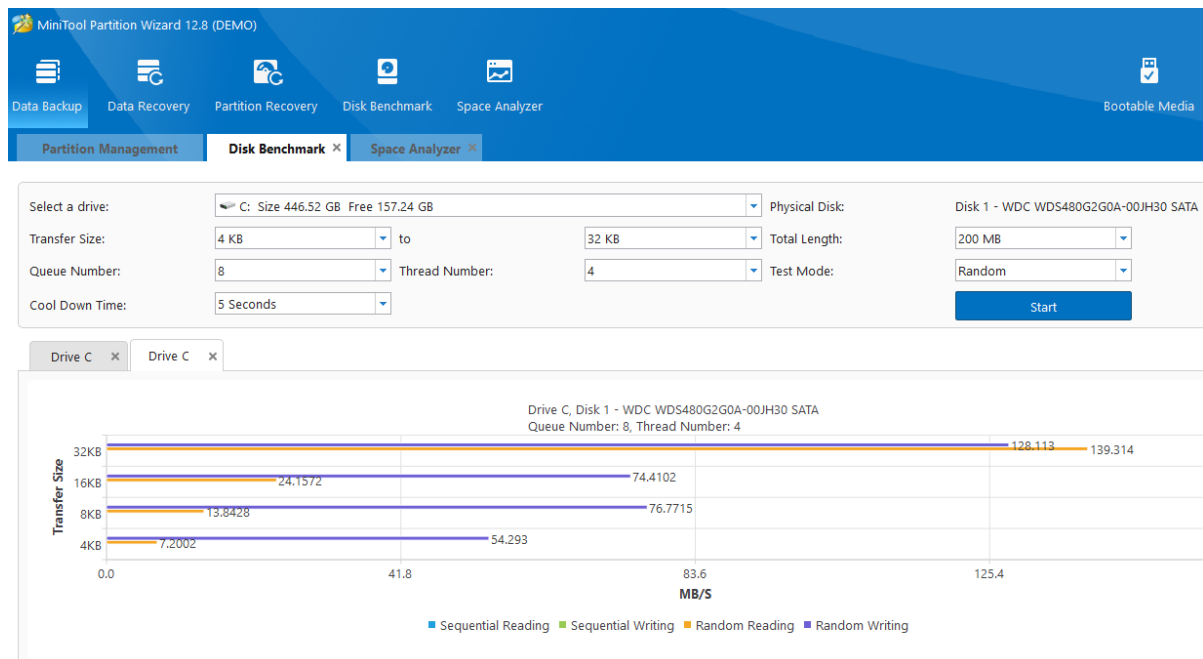


Figure A6: Example of Disk Benchmark (Test Mode: Random)